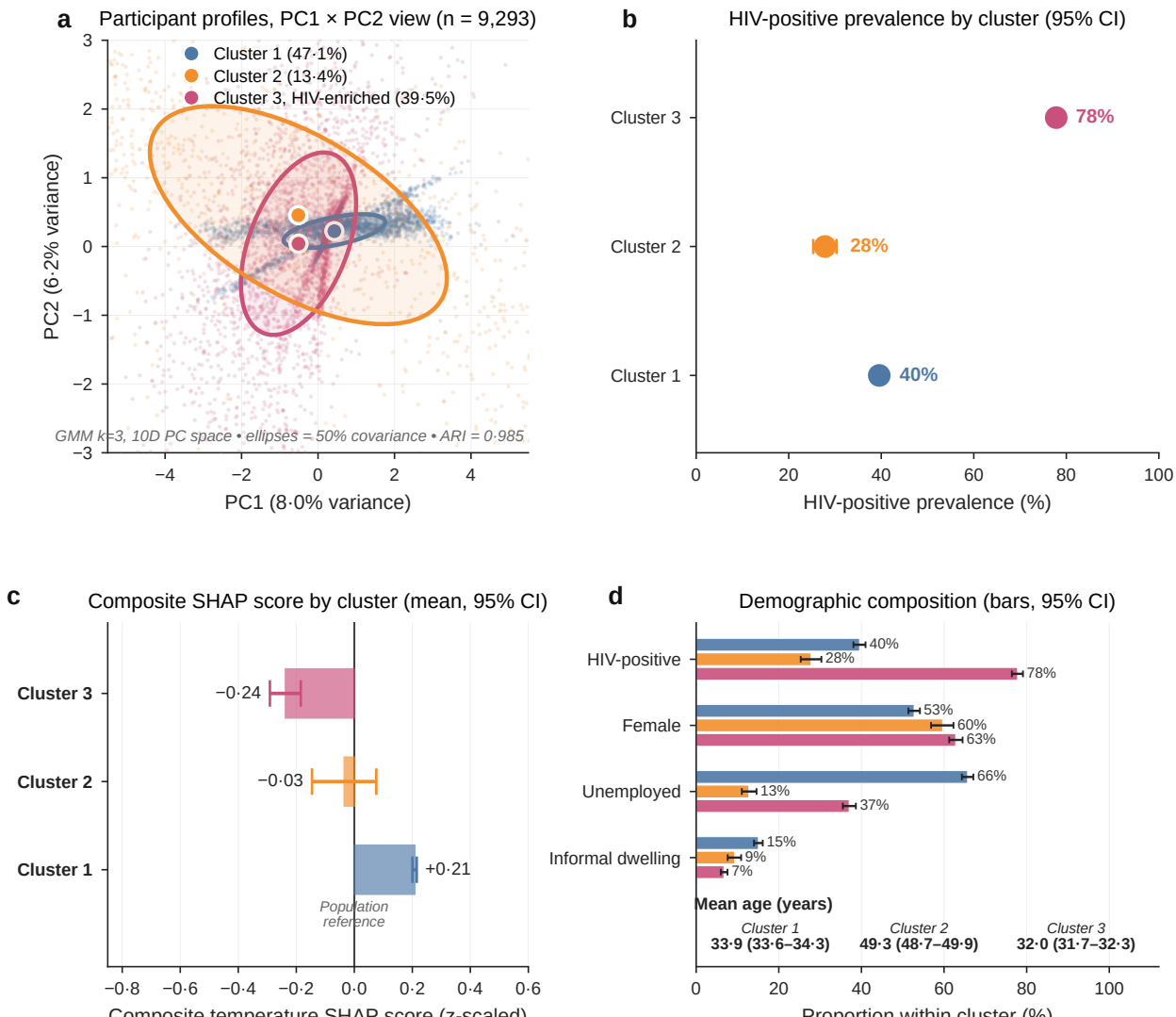

a
Participant profiles, PC1 × PC2 view (n = 9,293)
Cluster 1 (47·1%)
Cluster 2 (13·4%)
Cluster 3, HIV-enriched (39·5%)
PC2 (6·2% variance)
PC1 (8·0% variance)
GMM k=3, 10D PC space • ellipses = 50% covariance • ARI = 0·985
b
HIV-positive prevalence by cluster (95% CI)
Cluster 3
Cluster 2
Cluster 1
78%
28%
40%
HIV-positive prevalence (%)
c
Composite SHAP score by cluster (mean, 95% CI)
Cluster 3
Cluster 2
Cluster 1
−0·24
−0·03
+0·21
Population reference
Composite temperature SHAP score (z-scaled)
d
Demographic composition (bars, 95% CI)
HIV-positive
Female
Unemployed
Informal dwelling
40%
28%
78%
53%
60%
63%
66%
13%
37%
15%
9%
7%
Mean age (years)
Cluster 1
33·9 (33·6–34·3)
Cluster 2
49·3 (48·7–49·9)
Cluster 3
32·0 (31·7–32·3)
Proportion within cluster (%)